HACKpack.

8 practical FDM techniques for cleaner, more reliable prints. All values are starting points only — adjust for your printer, nozzle, material, part geometry and slicer version. Techniques are written to stay usable across Bambu Studio, Orca Slicer, PrusaSlicer, Cura and similar FDM slicers. Setting names and menu locations may differ, but the underlying print principles stay the same.

8
TECHNIQUES

Bambu · Orca · Prusa · Cura
SLICERS

FDM
FOCUS

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8 Techniques.

Z-SEAM

Z-Seam Reduction with Scarf Joint

Reducing seam visibility on curved and organic geometries.

THE PROBLEM

Every FDM layer has a start and end point. When those transitions stack in one place, they create a visible seam line along the outer wall. On parts with corners, the seam can often be hidden. On curved or organic surfaces, it usually stays visible regardless of placement.

THE FIX

Use a scarf-style seam transition if your slicer supports it. Instead of ending and restarting extrusion abruptly, the slicer feathers the transition so the seam becomes less noticeable. This is most useful on round and curved surfaces. On sharp-edged parts, standard seam placement can work just as well.

STARTING POINTS

Scarf Seam Type	**Contour or Contour and Hole, geometry dependent**
Smart Scarf Seam	**Enable as a starting point**
Scarf Joint Flow Ratio	**Start around 0.8, adjust per material and nozzle**
Wipe Features	**Test carefully, can interfere with scarf in some slicer versions**

PRO TIP

Test scarf seam on a representative piece of geometry before applying to a full print. On flat, straight-walled parts, standard aligned or corner seam placement often gives equivalent results with less tuning.

SLICER NOTE

Slicer naming varies. In Orca and Bambu Studio, look for Scarf Joint Seam under the seam or perimeter settings. In PrusaSlicer, the concept may appear as seam blending or custom seam controls. In Cura, this feature may not be available natively. Match the principle, not the label.

THREAD TOLERANCES

Thread Tolerances

A calibration workflow to improve the fit of printed threads.

THE PROBLEM

Printed threads often come out tighter than the CAD model suggests. On internal threads and close-fit holes, dimensional accuracy is influenced by material flow, line width and extrusion behaviour, so the female side usually ends up too tight.

THE FIX

Keep the male thread at nominal size in most cases. Add a small compensation to the female side, then verify with a short test print before using the value on production parts. The correct offset depends on your printer, nozzle diameter, material and extrusion temperature.

STARTING POINTS

Female-side compensation	**+0.1 to +0.2 mm starting range, calibrate per setup**
Male side	**Keep at nominal, usually no compensation needed**
Thread profile	**ISO 60 degrees, reliable for FDM**
Minimum pitch	**Around 1.0 mm for consistent engagement**
Infill for threads	**Adequate walls and infill for strength**

PRO TIP

Calibrate with a short test print, one matching bolt and nut takes under 10 minutes. Document your offset per material, nozzle size and printer. A well-calibrated value can be reused across projects without repeating the process.

SLICER NOTE

The setting name varies by slicer. Look for XY Size Compensation, Hole Expansion or First Layer Compensation depending on your tool. In Orca and Bambu Studio, check the Quality or Advanced tab. The workflow is the same regardless of where the setting lives.

ELEPHANT FOOT

Elephant Foot — First Layer Accuracy

Improving dimensional accuracy at the base of a print.

THE PROBLEM

The first layer is pressed against the build plate and stays warmer longer than later layers. That can make the bottom perimeter spread outward and create a flare at the base of the part. Decorative models may tolerate this, but fit-critical parts often do not.

THE FIX

First correct Z-offset and bed levelling. If the first layer is too compressed, slicer compensation alone will not solve the root cause. Once the first layer looks right, use elephant foot compensation as a fine adjustment. A small chamfer on the model (0.3 to 0.5 mm) can also help visually and functionally.

STARTING POINTS

First step	**Calibrate Z-offset and bed levelling**
Elephant Foot Comp.	**Start around 0.1 to 0.2 mm where available**
First Layer Height	**Around 0.2 mm as a baseline**
First Layer Speed	**30 mm/s or below for better control**

PRO TIP

Get the first layer right visually before enabling any compensation. A first layer that looks crushed or translucent means the Z-offset is too low. Fix that first. Slicer compensation is a fine adjustment for an already correct first layer, not a workaround for poor calibration.

SLICER NOTE

Slicer naming varies. Bambu Studio and Orca use Elephant Foot Compensation directly. PrusaSlicer uses Elephant Foot Compensation similarly. Cura calls the equivalent setting Initial Layer Horizontal Expansion, entered as a negative value. The concept is the same across all of them.

IRONING

Ironing on Flat Top Surfaces

Reducing surface ridges on flat horizontal faces.

THE PROBLEM

Top layers are printed in rows, and those rows leave small ridges. Under raking light or in product photos, the surface can look uneven, especially with lighter materials or larger layer heights.

THE FIX

Enable ironing only on flat horizontal top faces. Ironing makes a low-flow finishing pass that reheats and flattens the ridges. It can improve appearance significantly, but it is less predictable on angled, curved or overhanging surfaces.

STARTING POINTS

Ironing mode	**Topmost Surface only**
Ironing flow	**Start around 10 to 15 percent, increase in small steps**
Ironing speed	**Start at 40 mm/s, reduce speed before raising flow**
When to skip	**Angled, curved or overhanging surfaces**

PRO TIP

Tune flow and speed as separate variables. Reduce speed first if the surface looks uneven. Only increase flow if ridges remain visible at reduced speed. Test on a sample piece per material. Optimal settings differ between colours and filament brands.

SLICER NOTE

Most major slicers support ironing under the same name. In Bambu Studio and Orca, look for Ironing under Quality settings. In PrusaSlicer, it is under Infill. In Cura, it is called Ironing under Shell settings. Always use the top-surface-only or topmost-surface option to avoid applying it where it can cause problems.

LAYER ORIENTATION

Layer Orientation and Part Strength

How print direction affects structural performance.

THE PROBLEM

FDM parts are anisotropic, not equally strong in every direction. The bond between layers is weaker than the material within a layer, so orientation can matter more than infill percentage for functional parts.

THE FIX

Before slicing, identify the main load path. Orient the part so the layer lines cross that load path instead of running parallel to it. A well-oriented part at moderate infill often outperforms a poorly oriented part at high infill.

STARTING POINTS

Infill pattern	**Gyroid or similar for multi-directional loads**
Infill	**40 percent or more for structural parts, orientation usually matters more**
Walls	**4 to 6 perimeters for functional parts**
Material	**PETG often offers better impact resistance and layer adhesion than PLA in many practical use cases**
Key principle	**Define load direction first, then set infill and orientation**

PRO TIP

Think about where the load is applied and in which direction before placing the part on the build plate. A bracket loaded in tension should have its layers crossing that tension. A hook printed flat is fundamentally weaker than the same hook printed upright, regardless of infill density.

SLICER NOTE

This principle is slicer-independent. No slicer setting changes the physics of layer adhesion. Use the slicer preview to confirm layer direction relative to the part, but the real decision happens before slicing.

FUZZY SKIN

Fuzzy Skin — Textured Outer Wall Finish

Adding a matte textured surface to outer walls.

THE PROBLEM

Smooth outer walls can look glossy, slippery and unfinished on some hand-held parts. They also tend to show layer lines and minor surface artefacts clearly.

THE FIX

Use a textured-wall feature to add controlled surface variation to outer walls. This can create a matte finish and improve grip. Results depend on printer, speed and material. Always test on a small sample piece before applying to a functional part.

STARTING POINTS

Apply to	**Outer walls only, not top or bottom surfaces**
Skin Thickness	**Start around 0.3 mm for a subtle effect**
Point Distance	**Start around 0.8 mm for a finer texture**
Test first	**Print a small block to evaluate the result before the full part**

PRO TIP

Print a test block and handle it before committing to a final part. The result you see on screen rarely matches what the printed texture feels like. A coarser texture improves grip but can affect tolerances on features that need to fit with other parts.

SLICER NOTE

Bambu Studio and Orca call this Fuzzy Skin. PrusaSlicer uses the same name. In Cura it may appear under Experimental or with similar naming. If the feature exists, the parameters to adjust are the same across slicers, even if the menu path differs.

COLD PULL

Cold Pull — Nozzle Cleaning

Removing residue and partial blockages from the nozzle.

THE PROBLEM

Nozzles rarely fail all at once. More often, partial buildup from burnt filament, pigment or previous materials causes inconsistent extrusion, surface defects or colour contamination.

THE FIX

Heat the hotend and purge loose material. Then lower the temperature to a filament-appropriate pull range and extract the filament in a slow, steady motion. Repeat until the pulled filament comes out clean. If it tears, increase temperature slightly. If it slides out without carrying debris, reduce it.

STARTING POINTS

Pull temp PLA	**Around 90 degrees C as a reference, varies by formulation**
Pull temp PETG	**Around 125 to 135 degrees C as a reference range**
Pull temp ABS	**Around 140 to 160 degrees C as a reference range**
Cleaning filament	**Nylon is often effective, adheres well to residue**
When to use	**After material changes, especially high to low temperature transitions**

PRO TIP

Pull slowly and steadily. A fast extraction is less likely to carry debris cleanly out of the bore. The extracted piece should show a smooth, tapered tip that mirrors the inside of the nozzle. Repeat the process until you get two or three clean pulls in a row.

SLICER NOTE

Cold pull is a printer maintenance method, not a slicer function. It applies identically regardless of which slicer you use. No slicer setting is involved.

REDUCING WARP

Reducing Warp in FDM Prints

Managing thermal deformation through combined measures.

THE PROBLEM

Warping happens because printed material contracts as it cools. The outer surfaces cool faster than the inner mass, creating stress that can lift corners and edges off the build plate. Slicer settings alone cannot fully eliminate this — it is fundamentally a thermal problem.

THE FIX

Address warp with multiple measures at the same time: improve bed adhesion, keep bed temperature stable, reduce thermal gradients and adjust part design where needed. A brim, a stable build environment and small geometry changes often work better together than any single setting.

STARTING POINTS

Brim width	**Around 8 to 12 mm for large flat parts, adjust to geometry**
Bed temp	**Keep consistent throughout, do not reduce after layer 1**
Enclosure	**Use one when material and printer allow, especially for warp-prone materials**
Draft shield	**Helpful for tall or thin parts in open environments**
Base geometry	**Small fillets or chamfers reduce stress concentration at corners**

PRO TIP

Treat warp reduction as a system problem, not a single-setting fix. Bed adhesion, enclosure temperature and part geometry all contribute. Correcting only one factor is rarely sufficient. Materials like ABS and ASA are more prone to warp than PLA and PETG under the same conditions.

SLICER NOTE

Warp reduction is mostly a printer and environment challenge rather than a slicer-specific one. Most slicers offer brim and draft shield options under similar names. The exact path may vary but the settings themselves are equivalent across Bambu Studio, Orca, PrusaSlicer and Cura.

You've got it. Now go build.

WHAT'S NEXT

NEXT GUIDE

Waterproof Enclosures

How to design, print and seal FDM enclosures that resist water ingress. PETG and ASA compared. Practical settings and test protocol included.

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